APPLIED CFD ON SMALL SCALE WIND TURBINE BLADES

Submitted by:

Harrison Tracy, Javanni Asis, Charles Hughes

AEROSPACE AND OCEAN ENGINEERING DEPARTMENT

VIRGINIA POLYTECHNIC INSTITUTE AND STATE UNIVERSITY

BLACKSBURG, VIRGINIA

12 MAY 2026

Introduction

The goal of this project was to develop a high-fidelity CFD framework to accurately capture repeatable wind turbine performance metrics such as thrust, torque, and power. The models were based on the current turbine design of the Virginia Tech Wind Turbine Team. Setting up a framework for CFD in wind turbine blade design was one of the contributing factors for the project. The following objectives were investigated through this project:

1. Develop an operational 3D, sliding mesh to capture the wake directly following a rotating small scale wind turbine
2. Compare performance metrics across three distinct turbine blade configurations
3. Investigate the impact of winglets on turbine performance under fixed inlet conditions

These objectives were achieved through the running of three separate simulations under the same boundary conditions, mesh configurations, and adjusted rotation rates in order to produce consistent results. The two main focuses for this study were on the implementation of winglets on small scale wind turbines and how the optimization of a turbine's blade geometry affects its overall importance.

Winglets have become industry standard in many areas of the aerospace field. The main industry which uses winglets is the fixed-wing aircraft field, where the winglets allow for a reduction in induced drag, and thus higher fuel efficiency. This is particularly useful for commercial and military aircraft, which may be subject to heavier loads, making higher fuel efficiencies very valuable. Winglets have also become increasingly common in the wind turbine industry over the last 30 years. Testing has been sparse on small scale wind turbines, however, which is the scale with which the Wind Turbine Team operates. Thus, this project provides a steppingstone for which CFD be compared with empirical data allowing for its use in future small scale wind turbine blade winglet development.

The effectiveness of blade design relies heavily on airfoil and blade geometry. Airfoils have been under development since the start of flight. The same is true for small-scale wind turbines. Every year, new airfoils are found, tested, and compared with previous iterations to determine whether improvements are made. The current testing techniques for small scale wind turbines include 3D modeling, printing, and wind tunnel testing. This process is long and tedious, for often little to no progress. The use of an accurate CFD model in this process would expedite the process of blade testing and thus improve the blade design as a whole.

Blade Configurations and Geometries

Three distinct, SG6043 type blade configurations were analyzed in this investigation: two with winglets, and one without. The SG6043 was selected as the primary blade for the Wind Turbine Team this year and thus used in each of our simulations. We hoped to have empirical data to compare simulations to, however, due to electronics issues during testing, torque and power data were unable to be collected from the physical turbine. The turbine blade without winglets is optimized for a 3.7 tip speed ratio (TSR). This blade was used as a baseline for further comparison with the other two blades. The second and third blade tested both included winglets but were optimized for different tip speed ratios, 3.7 and 4.1 TSR respectively. Tip speed ratio is one of the biggest factors in a wind turbine's cut in speed, the wind speed at which the blades start rotating. Therefore, the TSR greatly determines the efficiency with which a wind turbine will operate. The specific TSR of 3.7 and 4.1 related to physical testing cut in values for the small-scale wind turbine, with cut in speeds of 4 m/s and 5 m/s respectively. All three configurations were expected to perform differently under the fixed inlet conditions due to the difference in TSR optimization and winglet application. In order to best assess each blade's performance, three parameters were measured: thrust, torque, and power. Probes were used in each simulation to monitor and plot both the thrust and torque. The power was measured by multiplying the measured torque value by 251. Each blade was made in SolidWorks along with a center configuration. The final blade configurations seen in **Figure 1** were made using SolidWorks assembly, with each blade being pitched 120 degrees, consistent with the pitching of the physical turbine at 10 m/s.



Figure 1: 4.1 TSR With Winglet (Left), 3.7 TSR No Winglet (Middle), and 3.7 TSR Winglet (Right)

Mesh and Simulation Set-Up

The blade geometries seen above in **Figure 1** were imported into StarCCM+ and a mesh was created around the blade origin. The mesh domain used across the three simulations was a cylindrical freestream with a dome-shaped inlet area, as seen outlined in **Figure 2**. The height of the domain and the radius of the inlet were both set to 10 blade lengths, or 2.5 m. The domain was extended 20 blade lengths behind the turbine, or 5 m. A rotational sliding mesh was generated in a thin, cylindrical domain localized around the blades. The rotating mesh was made to extend past each winglet/wing tip by 3 mm, seen in **Figure 3**. A refinement cylinder was created and extended backwards 5 blade lengths, or 1.25 m behind the turbine. This region was used to refine the wake area behind the turbine. The radius of the cylinder was set to 1.5 blade lengths, or 0.375 m in order to capture the tip vortices in the higher refined region. This region can be seen as the darkest black region in **Figure 2**. Initially, this region was set to 1 blade length, or 0.25 m, however the results needed further wake improvement, therefore the wake cylinder was extended to 0.375 m.

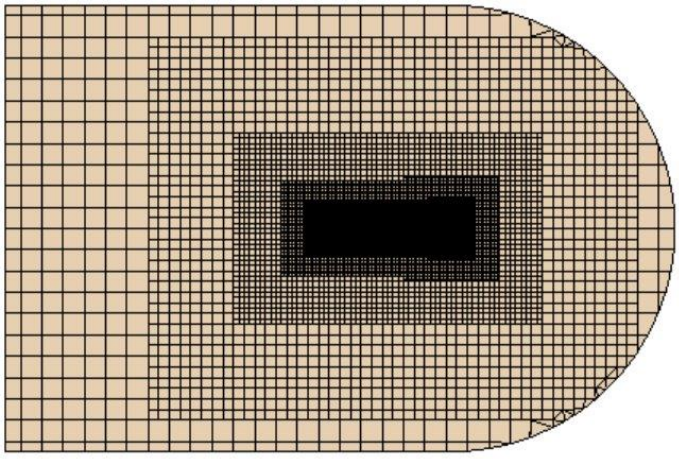


Figure 2: Mesh of the Total Computational Domain

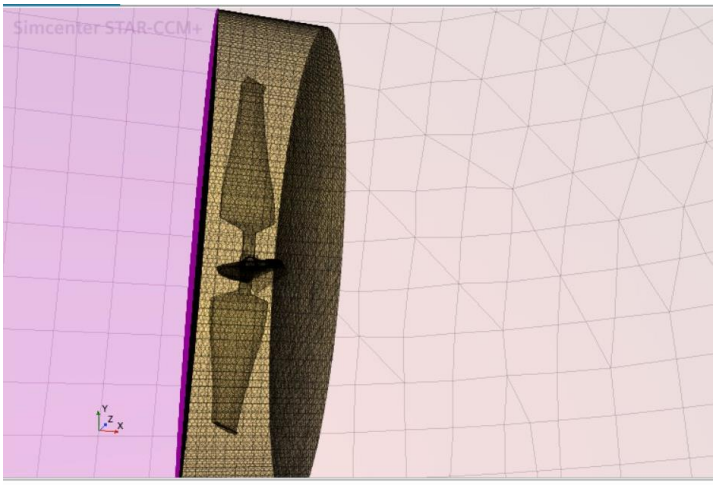


Figure 3: Zoomed-in Scene of the Refined Rotation Mesh Around the blades

Each boundary region was created and generated prior to the start of mesh fabrication. Two new regions were also created using the subtract feature. The rotating region was subtracted from the entire domain, creating a region which could mesh everything that would not be rotating. The second new region was subtracting the blade from the rotation region, creating a region separating the blades from the rotation. The entire mesh setup flow chart can be found in **Figure 4**.

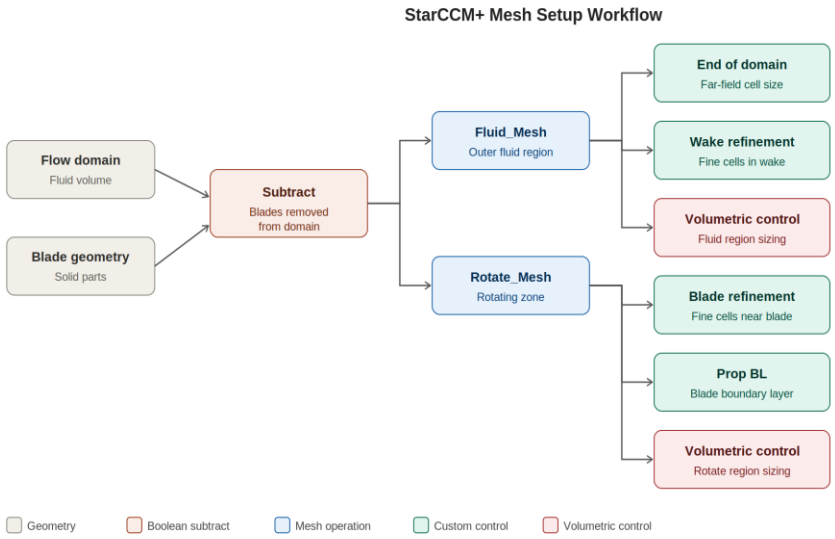


Figure 4: Simulation Mesh Workflow

Firstly, the blade geometries are subtracted from the rotating subdomain to enforce wall boundary conditions without geometrical interference. Next, two regions are generated: the outer fluid domain containing the inlet, and outlet, and an inner rotating refinement area. Various surface and volumetric controls are implemented in these two domains to refine the wake, blade surfaces, propeller boundary layer, and volumetric region growth rates. The selected base size was one blade length, and all other mesh dimensions were specified as percentages of this base size. The total mesh size configuration is summarized in the following table:

Parameter	Value
Base Size	0.2 m
Target Surface	10%
Minimum Surface	1%
End of Domain Cell Size	43%
Rotating Region Size	1%
Default Volumetric Control	1.2%
Prop BL Target Surface	0.3%
Prop BL Minimum Surface	0.15%
Propeller BL Surface Curvature	150 pts/circle, max 200
Rotating Mesh Polyhedral Mesher	0.8%

Table 1: Mesh Parameters

Several mesh iterations were tested during these simulations, with the base size, and Prop BL values being changed the most between simulations. The above table displays the final values for the mesh used in all three simulations.

Physics Models and Simulation Set-Up

In order to best measure the simulated wind turbine performance, certain physics models and initial conditions were implemented to obtain accurate results. The simulations were 3D, due to the sliding mesh, and used an implicit unsteady solver to capture the time-varying flow. The implicit unsteady condition is necessary due to the highly unstable wake behind the rotating turbine. Additionally, a coupled convection scheme with a K-Omega turbulence model was used to capture turbulence and compressibility effects. It was determined that this combination best tracked the highly complex turbulence produced in these simulations when compared to other models. The initial conditions chosen were based on a standardized blade tip speed ratio of 6 for the given blade lengths. This standardized ratio determined the freestream inlet velocity as well as the blade rotation rate. In addition, the time step was determined based on the rotation rate to have the turbine rotate 3 degrees per timestep. Upon further review, it was determined that this step size may be too big to record data accurately. The full physics model selections and initial conditions are summarized in the table below:

Parameter	Value
Solver	3D Implicit Unsteady
Convection Scheme	Coupled
Turbulence Model	K-Omega
Time Step	1.66e-4
Inner Iterations	5
Inlet Mach Number	0.029
Inlet Temperature	300K
Inlet Pressure	0 Pa
Initial Velocity	10 m/s
Initial Pressure	0 Pa
Static Pressure	300K
Reference Pressure	1 atm

Table 2: Simulation Parameters

Results and Discussion

The following three simulations were run using the mesh configuration and boundary conditions listed in the above sections.

A. 3.7 TSR with no Winglets

The simulation with blades designed for 3.7 TSR with no winglets was used as a baseline for the following two simulations. The simulation ran for roughly 1250 iterations over 6 hours in Arc. The residuals stayed relatively constant, dropping between 0.5 to 0.001. Although the residuals dropped more than in the other simulations, as seen in A1 of the appendix, they do not necessarily confirm that the simulations are accurate. Further testing with more refined regions will be needed to confirm or bring doubt to the validity of the simulation. The torque and thrust monitors became stable after roughly 500 iterations. The time-averaged power was found to be 194.8 W, which is compared to the other simulations in the next section. The simulation also produced oscillating tip vortices in the wake behind the turbine, which can be seen below in **Figure 5**.

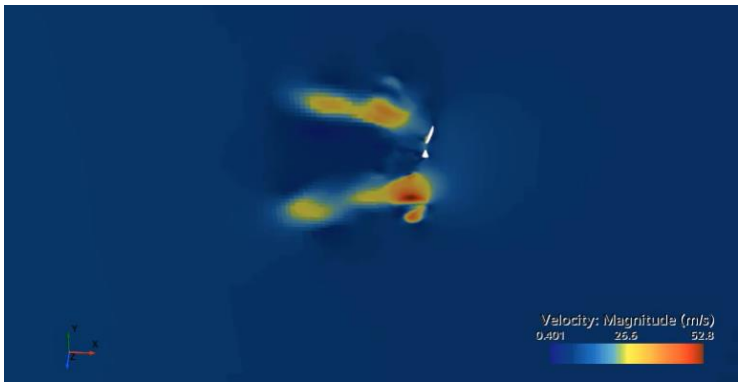


Figure 6: 3.7 TSR Simulation State Velocity Field

B. 3.7 TSR with Winglets

The 3.7 TSR blades with winglets produced a stable thrust and torque response after the initial transient, with the power monitor giving an average value of 719.12 W over the last 5 rotations. The velocity field showed a developed wake behind the rotor and visible tip vortex structures near the blade tips suggesting the winglets affected the downstream flow as shown in **Figure 6**. However, the residual history showed that the turbulent kinetic energy residual continued to grow instead of reaching a stable oscillatory pattern as shown in **Appendix A3**. Because of this, the case was not considered fully converged. The thrust, torque, and power monitors were still useful for comparing the blade design trends, but due to the elevated TKE residual means the results from this simulations should be interpreted with caution.

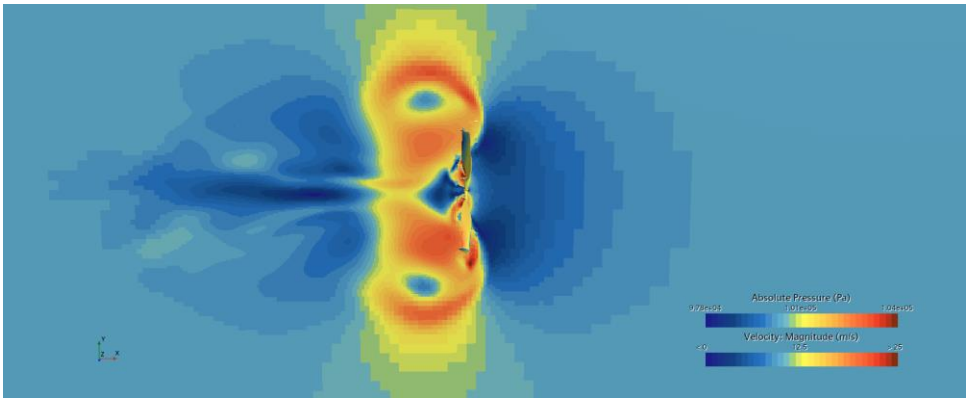


Figure 6: 3.7 TSR with Winglets Simulation State Velocity Field

C. 4.1 TSR with Winglets

Of the three simulations run, the 4.1 TSR with winglets appeared to have the most stable results. The simulation ran out for about 13,000 iterations for a total elapsed time of roughly 10 hours. The residuals stayed consistently stable and dropped every inner iteration throughout the duration of the run (**Appendix A3**). The torque, thrust, and power monitors were stable, with periodic oscillations occurring at each blade's rotation as expected in **Figure 7**. The measured time-averaged power was 904.403 W, and the simulation generated a physically consistent flow field capturing tip vortices and wake formation over time. The final state velocity magnitude field is shown below:

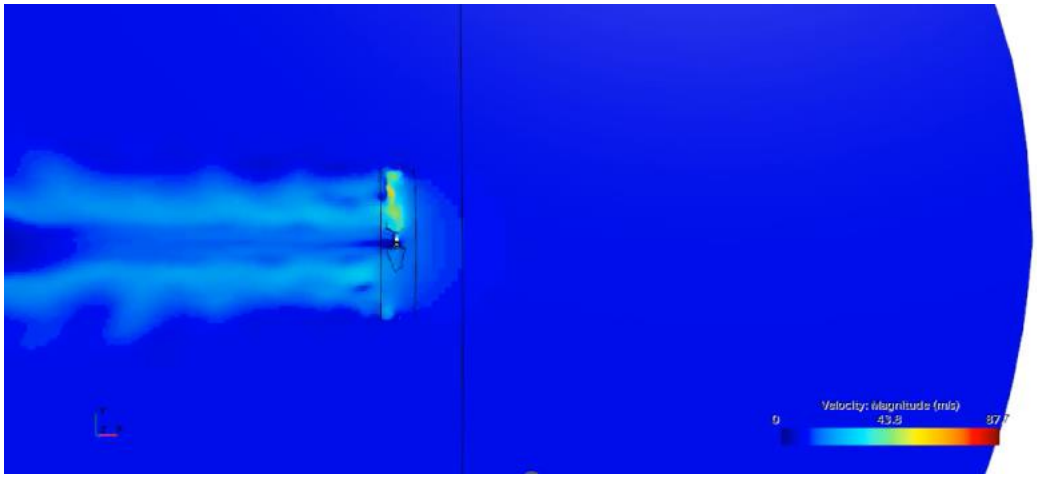


Figure 7: 4.1 TSR Final State Velocity Field

D. Result Comparison

The final torque, thrust, and power values obtained from the three simulations are summarized in the following table:

Blade Type	Thrust (N)	Torque (N-m)	Estimated Power (W)
3.7 TSR, No Winglet	26.275	0.77	194.8
3.7 TSR, Winglet	9.969	3.598	904.4
4.1 TSR, Winglet	22.889	2.862	719.12

Table 3: Simulation Results

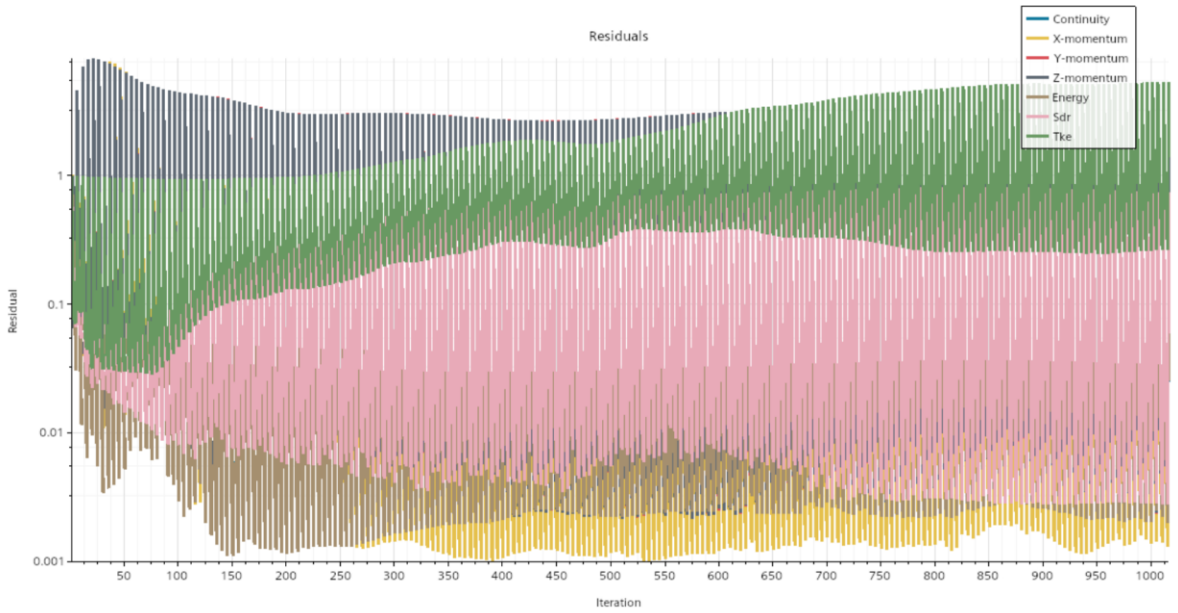
Upon inspection of the final values, there are clear inconsistencies and variations within the measurements. In a real-world scenario, if these three geometries were subject to the same conditions and had the same blade lengths, they would perform very similarly to each other, within 5-10 percent. Given the large differences in the measured values, it can be concluded that it's unlikely for all three simulations to be physically accurate. This indicates that there was a major error in the simulation set-up or sliding mesh generation. Further simulations should be run in order to determine which, if any, of the current simulations are accurate.

Future Work

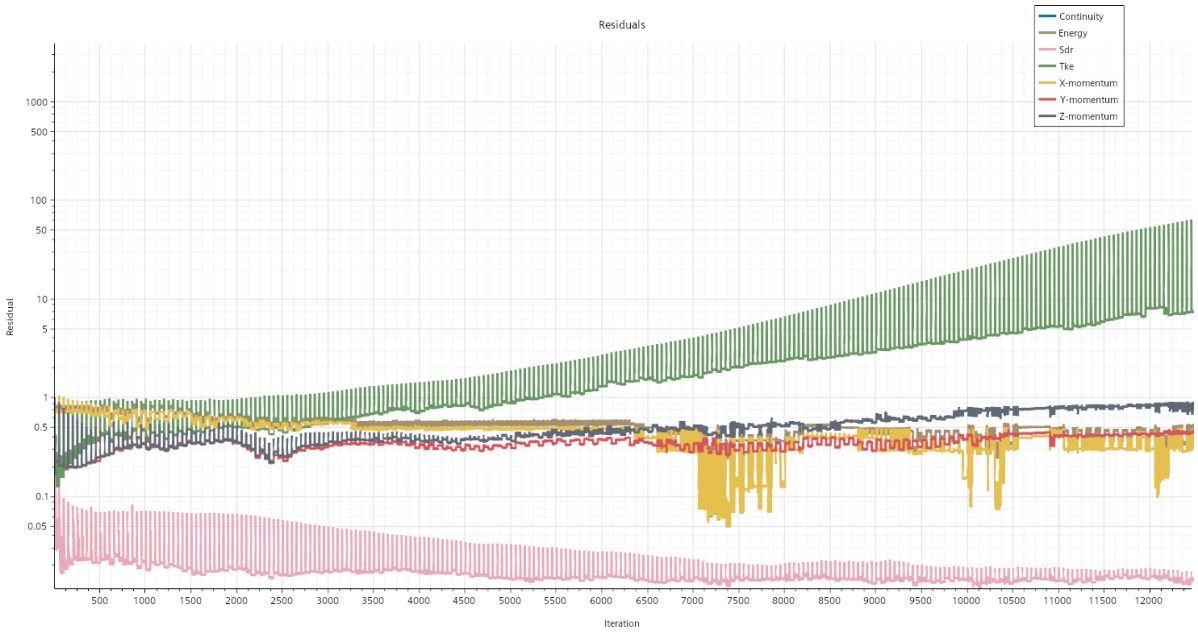
This report sets out a good baseline for future work in the wind turbine field using CFD. The next steps include further refinement of simulations, focusing on decreasing residuals to a reasonable value, comparison with experimental results, and thorough justification for assumptions made. In terms of the current simulation, once results are verified, the next step would be to investigate the tip vortices from the blades and compare between winglet and wingless cases. Since this is one of the main justifications to the winglet addition, it would be interesting to look at how much they actually affect wing tip vortices. Another potential path forward is to look at the effects of

turbine yaw and pitch orientation on power generation and efficiency. In real world application, the wind is not always hitting the turbine head on, therefore it is important to improve the turbine performance at angles in which the wind is coming in at an angle as well as head on.

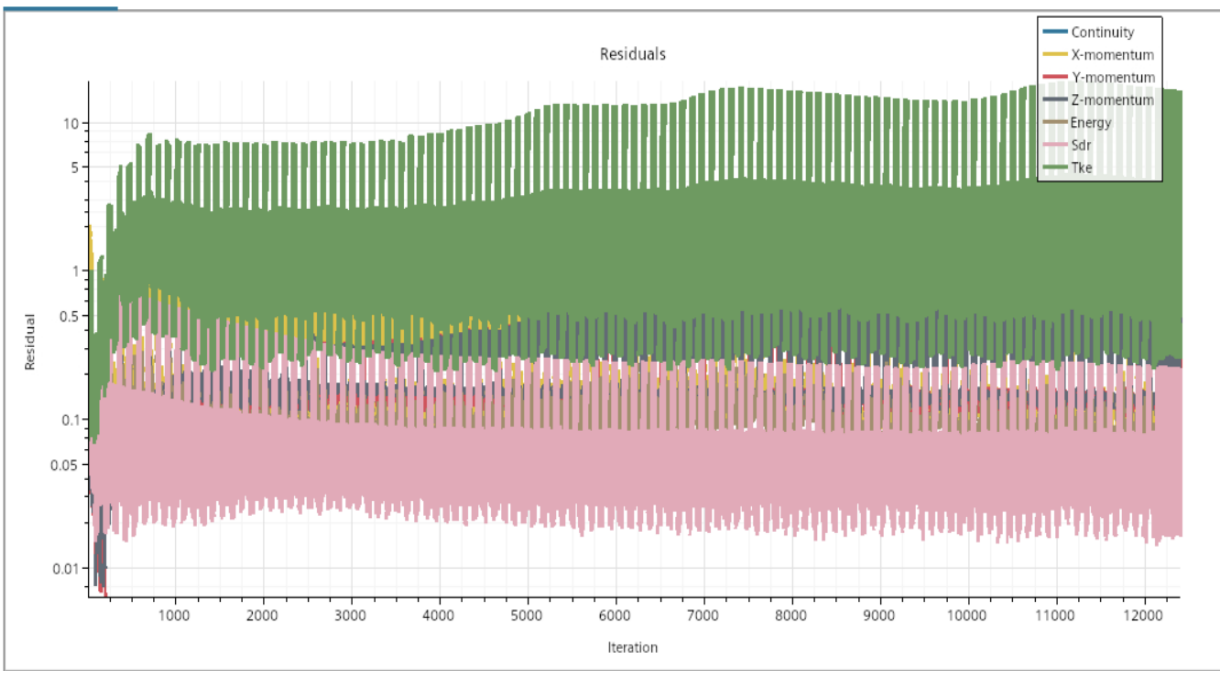
Appendix



A1: Residuals from Simulation of 3.7 TSR and No Winglet



A2: Residuals from Simulation of 3.7 TSR With Winglet



A3: Residuals from Simulation of 4.1 TSR With Winglet